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Number of questions	5
Number of pages	3

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UNIVERSITY OF SRI JAYEWARDENEPURA
FACULTY OF TECHNOLOGY

Bachelor of Information and Communication Technology Honours Degree

Academic Year 2022/23

Fourth Year Second Semester End Examination – March 2025

ITC 4172 Entrepreneurship and Small Business Management

Duration: Two (2) Hours

Read and follow the instructions given below.

- This paper contains five (5) questions on three (3) pages.
- “Index number” should be written on top of each page of the answer script, and pages must be numbered appropriately.
- Answer all questions.
- Start each question on a new page.
- Draw clearly labelled diagrams wherever necessary.
- Non-programmable calculators could be used.
- If you have any doubts as to the interpretation of the wording of a question, make your own assumptions, but clearly state them in the answer script.

Question 1**(Total Marks 20)**

- a). All entrepreneurs are business individuals, but not all business individuals are entrepreneurs.

State the key differences between an entrepreneur and a business individual, with reference to their nature, goals, risk tolerance, market positions, and orientation. (10 Marks)

- b). Provide a real-world example of a successful entrepreneur and briefly explain how they went through the entrepreneurial process. (10 Marks)

Question 2**(Total Marks 20)**

- a). Select an industry of your choice (e.g., manufacturing, apparel, or IT) and apply Porter's Five Forces Model to assess its attractiveness to start a new business. Explain each force and its impact on the profitability of the industry. (10 Marks)

- b). Analyze the unique marketing challenges faced by small businesses in Sri Lanka, focusing on segmenting, targeting, and positioning (STP), as well as the marketing mix (4Ps), and provide relevant examples to support your analysis. (10 Marks)

Question 3**(Total Marks 20)**

- a). Examine the significance of conducting the four (04) main types of feasibility studies when launching a new business venture. Provide relevant examples to support your evaluation.

target market
organizational
financial

(8 Marks)

- b). Evaluate and justify a suitable business model that can be applied to achieve a competitive advantage in a highly competitive (red ocean) market. Support your evaluation with a relevant example. (12 Marks)

Question 4**(Total Marks 20)**

- a). Critically evaluate the differences between sole proprietorships, partnerships, and limited liability companies (LLCs). In your answer, include how each legal structure impacts business operations, decision-making, and liability management, providing examples from both local and international businesses to illustrate how these structures influence business success and sustainability. (9 Marks)

b). Develop four (04) strategic solutions, with relevant examples, to address the key challenges contributing to the high failure rate of Small and Medium Enterprises (SMEs) in Sri Lanka, including:

- I. Poor sales → Ansoff matrix
- II. Inappropriate product portfolios → BCG Matrix
- III. Lack of strategic planning after reaching the maturity stage → BPR
- IV. Intense market competition → Porter's strategic solutions

(11 Marks)

Question 5

(Total Marks 20)

Evaluate the importance of the following concepts with industry-level examples.

- a) Mergers & acquisitions
- b) Technology transfer
- c) Digitalization of business
- d) Downsides of technology
- e) Intellectual property rights

(4 Marks x 5 concepts = 20 Marks)

*****End of the question paper*****

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Number of pages	04
Number of questions	05



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Academic Year 2022/2023

Fourth Year Second Semester End Examination – February / March 2025

ITS4243 Microservices and Cloud Computing

Duration: Three (03) Hours

Read and follow the instructions given below:

- Answer all the questions.
- “Index number” and course code “ITS4243” should be written on top of each page of the answer script, and pages must be numbered appropriately.

Question 1

(Total 15 Marks)

- a) Briefly explain your understanding of Software Architectural Style. (03 marks)
- b) "Microservices design patterns are crucial because they address the challenges that arise when building complex systems." (06 marks)
 - i. Define a design pattern.
 - ii. List down two (2) microservices design ^{patterns} ~~principles~~ and explain the application of each of them.
- c) Describe three (3) cloud deployment models. (06 marks)

Question 2

(Total 15 Marks)

Explain the following each of the concepts in the context of Microservices and Cloud Computing.

- a) Kafka. (05 marks)
- b) SAAS, PAAS, IAASx. (05 marks)
- c) Reserved instances, Right sizing. (05 marks)

Question 3

(Total 25 Marks)

- a) Explain two (2) methods that can be used to define the scope and right-size microservices. (08 marks)
- b) Describe three (3) technologies used in cloud native development. (09 marks)
- c) Observe the code below and explain it using your java vocabulary. (08 marks)

```
import java.sql.*;
public class Postgreconn {
    public static void main(String[] args) throws SQLException {
        int student_id=3;
        String student_name="Harsha";
        int student_marks=70;
        String url = "jdbc:postgresql://localhost:5432/demo";
        String uname = "postgres";
```

studentId = 3
name = Harsha
marks = 70
url =
uname =

```

String pass = "oracle";
System.out.println("Connection Established");
String sql = "insert into student values (?, ?, ?)";
Connection con =
DriverManager.getConnection(url,uname,pass);
PreparedStatement st = con.prepareStatement(sql);
st.setInt(1,student_id);
st.setString(2,student_name);
st.setInt(3,student_marks);
st.execute();
con.close();
System.out.println("Connection Closed");
}
}

```

Answer the Questions 4, and 5 based on the use case below.

Use case: Developing a Migration Strategy from Service oriented architecture to microservices architecture for Real-Taste-Pi.

Real-Taste-Pi is a multinational chain of fast-food restaurants. Like many fast-food chains, they operate a franchising model where, as the franchiser they license their know-how, procedures, intellectual property, use of their business model, brand and rights to sell their branded products and services to individual franchisees. In return, the franchisee pays fees and agrees to comply with certain obligations. Real-Taste-Pi was one of the original fast-food outlets to establish in the UK. Throughout the 1970s & 1980s it was extremely successful with over 1,000 outlets, but other competitors have encroached on their market share and now it only has a few hundred outlets mainly due to lack of technological adaptation. However, a new investment company named OSIC (Old Stuff is Cool) is hoping to revitalize the Real-Taste-Pi brand by trading on its heritage status and the new popularity of the retro dining experience, in the style of e.g. 1940s tea rooms. Real-Taste-Pi originally had strict guidelines on how different aspects of the franchises business should be set up, particularly the style, branding, menu and other 'visible' front-of-house features. However, it permitted some degree of choice with the back-office systems to enable franchisees to have some flexibility and thus negotiate a better deal. Therefore, outlets currently operate different types of Point-of-Sales (PoS) systems. These systems collect a variety of different types of information, depending on their age.

Real-Taste-Pi currently using monolith software architecture as their application design. They are looking for OSIC to get the app migrated to microservices software architectural style. But OSIC is looking for some advice on a specific strategy for the migration of a system within the context of Real-Taste-Pi.

Question 4

(Total 25 Marks)

Suppose that you have appointed as the new Solution Architect to this project, and you are supposed to direct “OSIC” to take the enterprise approach to develop the solution.

- a) Briefly explain what a requirement and a user story is. (05 marks)
- b) Explain the enterprise software development approach. (07 marks)
- c) Describe the important areas to consider under Business Architecture. (08 marks)
- d) Explain the importance of having a business case document. (05 marks)

Question 5

(Total 20 Marks)

As a Solution Architect to this project, you are supposed to assist “OSIC” to decide upon the software architecture style.

- a) Explain how you justify microservices as the suitable architecture for this solution. (05 marks)
- b) Describe the design patterns that you are suggesting for this solution. (07 marks)
- c) Create a solution architecture diagram showing microservice communication, including a discovery server. (08 marks)

*** End of the Paper***

Signature

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Academic Year 2022/2023

Fourth Year Second Semester Examination – February/March 2025

ITC4182 Professional Practices in ICT

Duration: Two (2) Hours

Read and follow the instructions given below:

- This paper contains four (4) questions in three (3) pages.
- Answer all questions.
- “Index number” should be written on top of each page of the answer script, and pages must be numbered appropriately.

Question 1

(Total 25 Marks)

- a) Differentiate professional practice and professional ethics. 3 Marks
- b) Describe two (02) professional organizations that ICT professionals in Sri Lanka can join, highlighting their purpose, benefits, and role in the industry. 4 Marks
- c) As ICT professionals, your primary focus is on developing systems and applications. However, strong verbal communication skills are equally important. 9 Marks
Examine three (03) situations where verbal communication is essential for ICT professionals.
- d) Collaborative tools are essential in the modern ICT project environment, and many advanced tools are available for this purpose. However, there are still several areas that could be improved to enhance their effectiveness. 9 Marks
As an ICT professional, propose three (03) key areas for improvement to make modern collaborative tools more effective.

Question 2

(Total 25 Marks)

- a) Explain the importance of team dynamics in the success of ICT projects. 3 Marks
- b) Assess three (03) characteristics of an *effective* leader. 6 Marks
- c) Critically analyze two (02) common leadership styles that are effective in the modern ICT setting. Discuss how each leadership style applies to ICT organizations and contributes to their overall success. 6 Marks
- d) Suppose you are starting an IT company that will handle multiple projects, including research and development, where meeting deadlines is crucial. Additionally, you plan to hire primarily interns to help manage costs. 10 Marks
 - i) Discuss the organizational structure (or combination of structures) you are implementing to achieve your goals. Justify your choice by discussing how it supports efficiency, collaboration, and innovation.
 - ii) Assess the organizational culture you aim to adopt to create an environment conducive to achieving both organizational and project targets. Highlight key cultural values, practices, and their impact on performance and teamwork.

Question 3

(Total 25 Marks)

- a) Explain Geographical Indications (GI) providing two(02) examples in the Sri Lankan context. 4 Marks
- b) Discuss the importance of trademarks in the ICT industry providing examples. 5 Marks
- c) Explain two (02) notable privacy and data protection cases worldwide that serve as exemplary decisions in the field. Discuss how these cases have influenced the development of data protection policies globally. 8 Marks
- d) With the assistance of relevant examples, critically analyze the effectiveness of Sri Lanka's legal and regulatory framework in protecting online privacy and ensuring data protection. 8 Marks

Question 4

(Total 25 Marks)

- a) Briefly explain the importance of *capacity to contract* in ICT-related Contracts. 3 Marks
- b) Evaluate the role of *consideration* and *Justa Causa* in the Sri Lankan contract law. 6 Marks
- c) Analyze the general structure of a contract and highlight the essential clauses that should be included. 6 Marks
- d) Assume you are drafting a contract for a freelance software developer hired to build a custom application for a client. Propose at least five (05) key aspects you would specifically include in the contract to ensure clarity, protect both parties, and define the scope of work. 10 Marks

**** End of the Paper*****

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UNIVERSITY OF SRI JAYEWARDENEPURA

Faculty of Technology

Bachelor of Information and Communication Technology Honours Degree

Academic Year 2022/2023

Fourth Year Second Semester Examination - February/March 2025

ITC4212: Information Systems Strategy and IT Governance

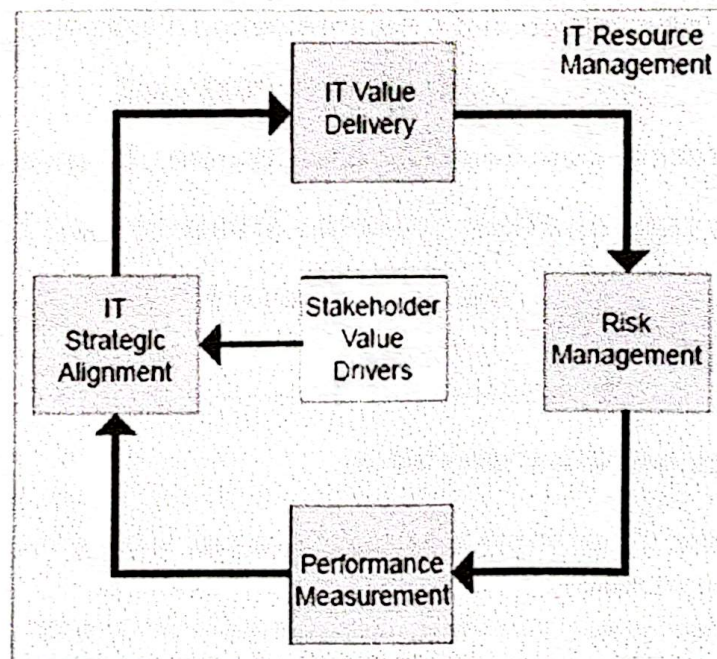
Duration: Two (2) Hours

Read and follow the instructions given below:

- 'Index number' should be written on top of each page of the answer script, and pages must be numbered appropriately.
- Marks allocated to each question are given alongside the question.
- Answers should be written in clear legible handwriting
- Answer all questions

Question 1**Total Marks 25**

- A. ✓ The ITG Institute defines IT governance as an integral component of enterprise or corporate governance. Explain and elaborate on this statement. (04 Marks)
- B. ✓ Top management plays a crucial role in the success of IT governance within organizations. Justify this statement with relevant explanations. (04 Marks)
- C. ✓ The following diagram is relevant to the focus areas of IT governance. (10 Marks)
Explain the role of each component (i.e., IT Value Delivery, IT Resource Management, Risk Management, Performance Measurement, and IT Strategic Alignment) in IT governance.



- D. ✓ Stakeholder value is a key outcome of effective IT governance. In reference to the above diagram (Question 1 C), critically evaluate how these components must be integrated to ensure long-term stakeholder benefits. (07 Marks)

Question 2**Total Marks 25**

- A. ✓ Describe the steps involved in implementing an IT governance framework in any organization of your choice. (08 Marks)
- B. ✓ Examine the dissimilarities between IT service management and IT governance in relation to the maturity of an organization's IT function. (05 Marks)
- C. ✓ Analyze the factors contributing to poor IT alignment. (06 Marks)
- D. A decrease in competitive advantage is a direct result of poor IT alignment. Justify this statement. (06 Marks)

Question 3**Total Marks 25**

- A. ✓ Business firms invest heavily in information systems to achieve strategic business objectives. Justify this statement with at least three (3) reasons. (06 Marks)
- B. Examine the role of environmental factors in information strategy planning and implementation. (05 Marks)
- C. ✓ Define Porter's Competitive Advantage Model. (06 Marks)
- D. ✓ Apply Porter's Competitive Advantage Model to the smartphone industry. Answer the question with appropriate examples from the industry. (08 Marks)

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Question 4

Total Marks 25

- A. Situational analysis, strategy formulation, and strategy implementation are (06 Marks)
the main IS strategy planning components. Define the role of each in IS
strategy planning.
- B. Discuss appropriate IS strategy for the following business strategies. (05 Marks)
- i. Low-cost producer
 - ii. High customer satisfaction
 - iii. Operate globally
 - iv. High quality-product
 - v. Quick-time to market
- C. Explain why communication is critical in IS strategy planning. (06 Marks)
- D. Apply the Managerial Levers Framework to an IT startup. How can it use (08 Marks)
organizational, control, and cultural variables to succeed?